# Cutting Flight Disruption

## A Data-Driven Travel Policy

Chart Champ Datathon

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# Agenda

The problem

What the data says

Recommendations

Demo: risk-scored booking

Pilot and future considerations



# The problem, and our answer up front

**9.74%**

of flights are major disruptions (cancelled, diverted, or 60+ min late)

**1 in 6**

flights arrives 30+ minutes late

same routes, same airlines:

early flights disrupt **4.94%**

late flights disrupt **14.48%**

**Our answer:**

- offer earlier departures
- set arrival buffers from measured delay exposure
- measure results before adopting a policy

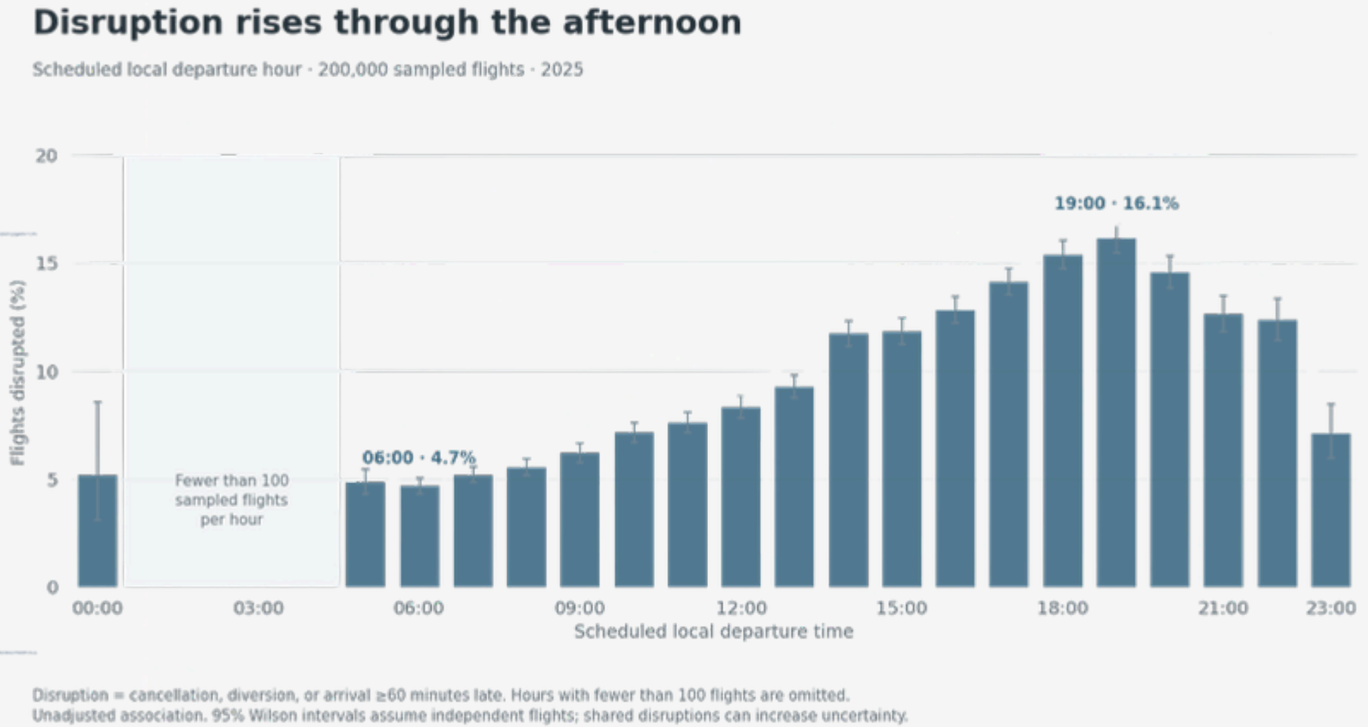


# EDA

## Disruption rises through the afternoon

**4.7%** → **16.1%**

at 06:00 at 19:00




Disruption = cancellation, diversion, or arrival ≥60 minutes late. Hours with fewer than 100 flights are omitted.
Unadjusted association. 95% Wilson intervals assume independent flights; shared disruptions can increase uncertainty.

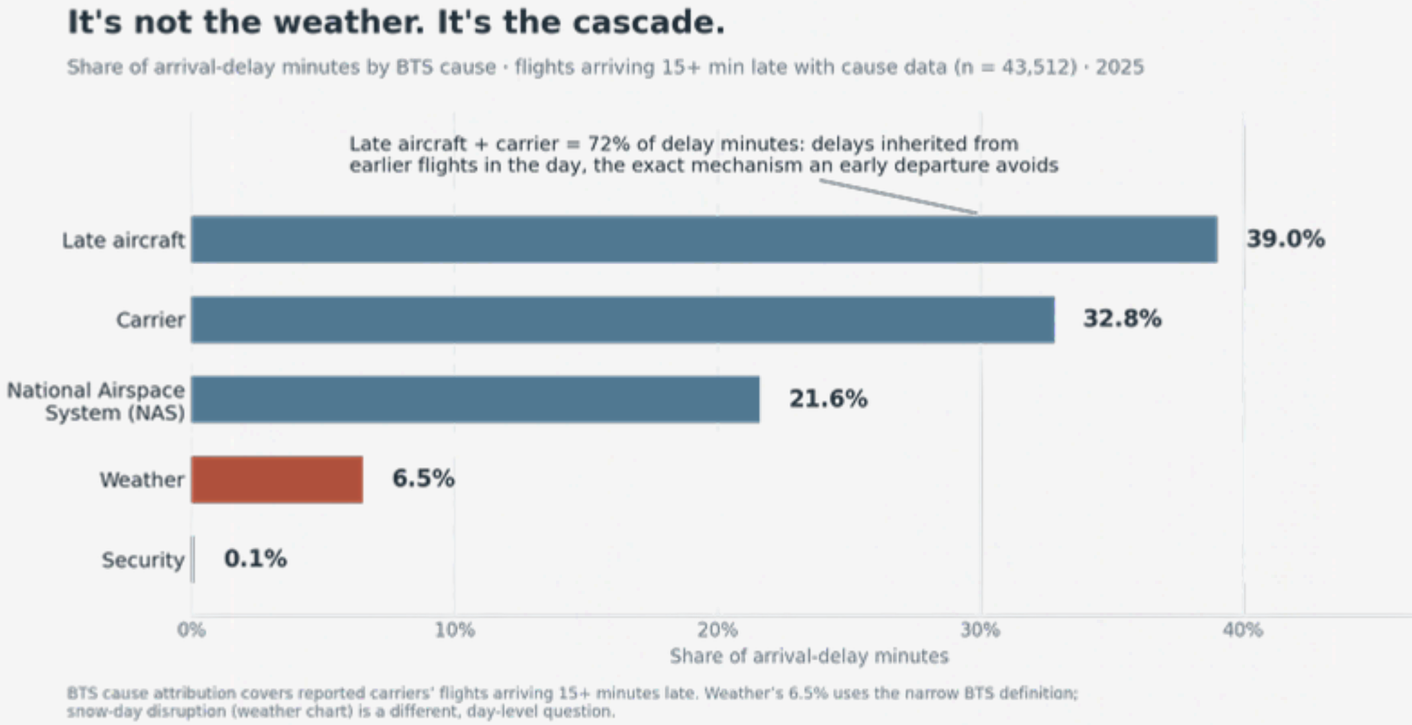
Morning departures are the safest flights of the day

Risk roughly triples by the evening peak

# It’s not the weather.
# It’s the cascade.

Delays are inherited from earlier flights in the day

That is the mechanism an early departure avoids




**39.0%**
late aircraft
of delay minutes

**32.8%**
carrier
of delay minutes

**6.5%**
weather
of delay minutes

# Price vs risk

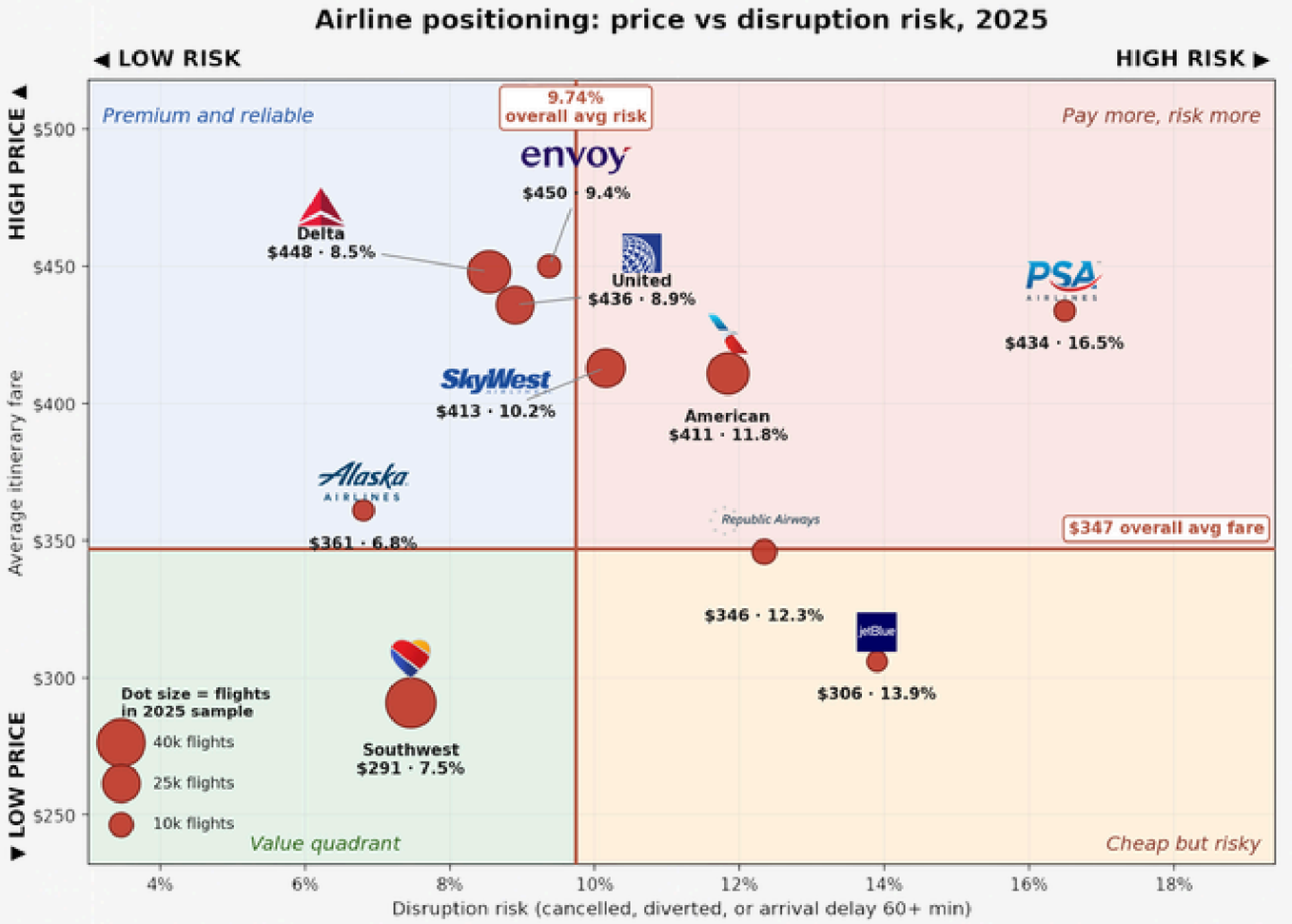
## you don’t always get what you pay for

**$347**
average fare

**9.74%**
average disruption

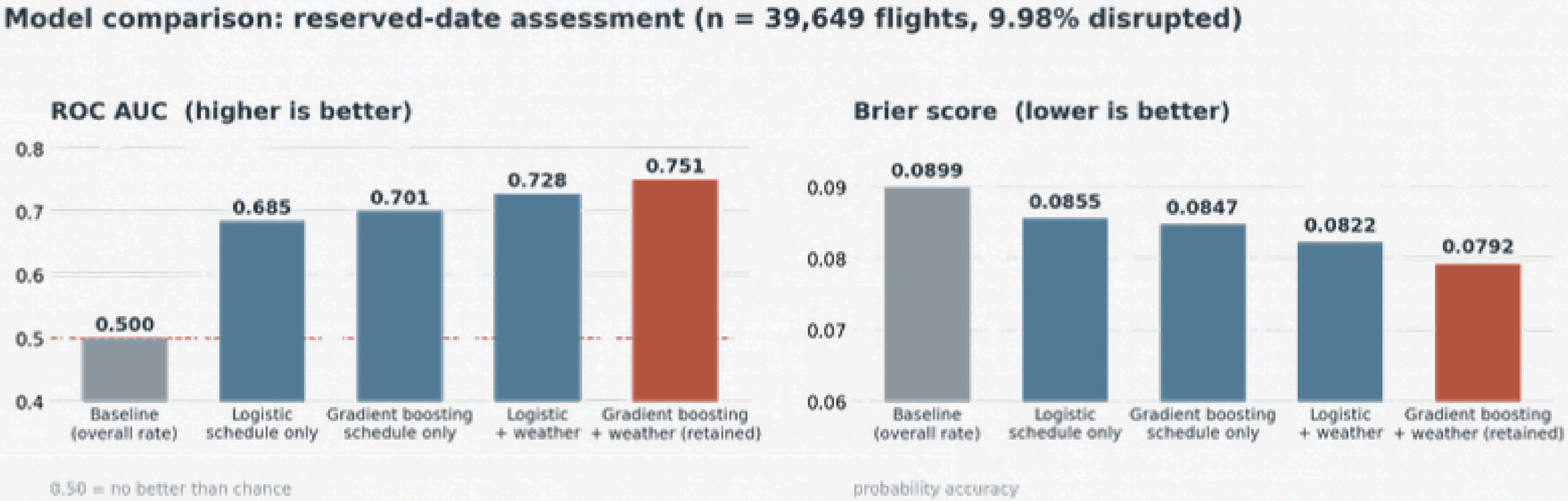
Some carriers charge premium fares for above-average disruption

Others are cheaper and more reliable at the same time




Fares: BTS DB1B ticket survey, 2025 H1, passenger-weighted mean itinerary fare. Risk: BTS On-Time 2025 sample; major disruption = cancelled, diverted, or arr. delay ≥ 60 min. Dot size: flights per carrier in the 200k sample. Benchmark cross: overall average fare $347 and overall disruption rate 9.74%. Logos via Google Flights / Wikimedia, for identification only.

# Model Comparison: What actually improved the score



# Recommendations

1. Offer earlier departures when suitable options exist: matched disruption 4.94% early vs 14.48% late across 2,227,943 flights

2. Use delay exposure to inform arrival-buffer discussions: 15.79% of flights miss a 30-minute allowance, 6.65% at 90 minutes, 3.30% at 180

3. Measure feasibility and outcomes before adopting a client policy: pilot on defined routes and record acceptance, disruption, deadline arrivals and actual costs






Link to Video

# Future considerations

## 01

Leg-level data: connections and missed connections were not accounted for, so nonstop-first is an assumption, not a finding

## 02

One year (2025), 10 selected carriers, and the sample overlaps the population: generalization beyond this setting is limited

## 03

The dataset has no booking alternatives, fares, meeting deadlines or disruption costs, so savings and real-world feasibility are still unproven



**Offer earlier departures.**

**Set arrival buffers from delay exposure.**

**Measure results before adopting a policy.**

Team 2

Questions welcome

